

Statistically Valid Inferences from Privacy Protected Data¹

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Google, 9/18/2020

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Solving Political Problems Technologically

Differential Privacy & Inferential Validity

A General Purpose, Statistically Valid DP Algorithm

The Algorithm in Practice

Convincing Facebook to Make Data Available

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Solving a Political Problem Technologically (via “constitutional design”)

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- [New Problem](#): **Sharing data without it leaving Facebook**

Data Sharing Regime $\leadsto$ Data Access Regime

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 - *no* uncertainty estimates

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Theories of Inference: Statistics vs. CS

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Population

:

Lindsay

Salil

Georgie

Gary

Meg

Abhradeep

Joshua

Annie

Bob

Ellen

Mean
income:

\$48

Quantity
of Interest

Theories of Inference: Statistics vs. CS

Population	Sample
:	X
Lindsay	✓
Salil	✓
Georgie	✓
Gary	✓
Meg	✓
Abhradeep	✓
Joshua	✓
Annie	✓
Bob	✓
Ellen	✓

Mean
income:

\$48

Quantity
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Theories of Inference: Statistics vs. CS

Population	Sample	\$
:	X	?
Lindsay	✓	122
Salil	✓	76
Georgie	✓	145
Gary	✓	96
Meg	✓	86
Abhradeep	✓	127
Joshua	✓	72
Annie	✓	132
Bob	✓	95
Ellen	✓	134

Mean
income:

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Classical
Inference

\$108

Quantity
of Interest

Usually
no direct
relevance

Theories of Inference: Statistics vs. CS

Population	Sample	\$
:	X	?
Lindsay	✓	122
Salil	✓	76
Georgie	✓	145
Gary	✓	96
Meg	✓	86
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Quantity
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Theories of Inference: Statistics vs. CS

Population	Sample	\$	+Privacy
:	X	?	
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Salil	✓	76	
Georgie	✓	145	
Gary	✓	96	
Meg	✓	86	
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Gary	✓	96		113
Meg	✓	86		125
Abhradeep	✓	127		97
Joshua	✓	72		101
Annie	✓	132		128
Bob	✓	95		83
Ellen	✓	134		201

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Query-
Response

\$111

Quantity
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Mean
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Differential Privacy and its Inferential Challenges

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- Estimators

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- Estimators
 - Classical Statistics: Apply statistic s to dataset D , $s(D)$

Differential Privacy and its Inferential Challenges

- Estimators

- **Classical Statistics:** Apply statistic s to dataset D , $s(D)$
- **DP Mechanism:** $M(s, D)$, with noise & censoring

Differential Privacy and its Inferential Challenges

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- **Statistical properties:** **usually biased**, **no uncertainty estimates**

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Solving Political Problems Technologically

Differential Privacy & Inferential Validity

A General Purpose, Statistically Valid DP Algorithm

The Algorithm in Practice

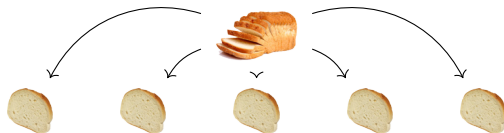
A Differentially Private Estimator

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Private data

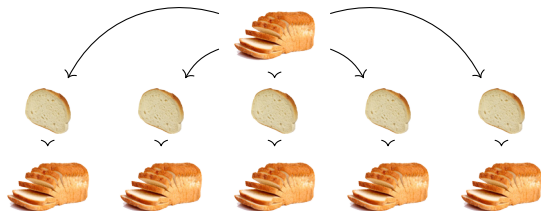
A Differentially Private Estimator



Private data

Partition

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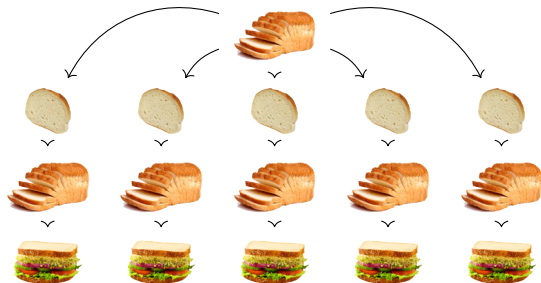


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Bag of little bootstraps

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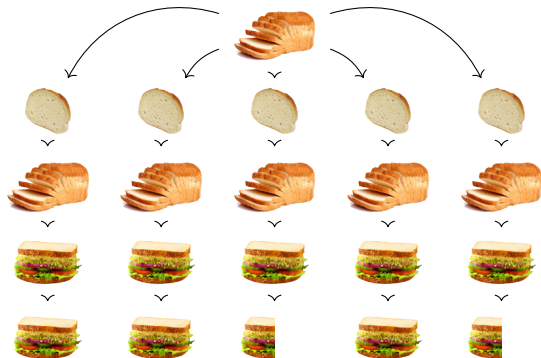
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Estimator

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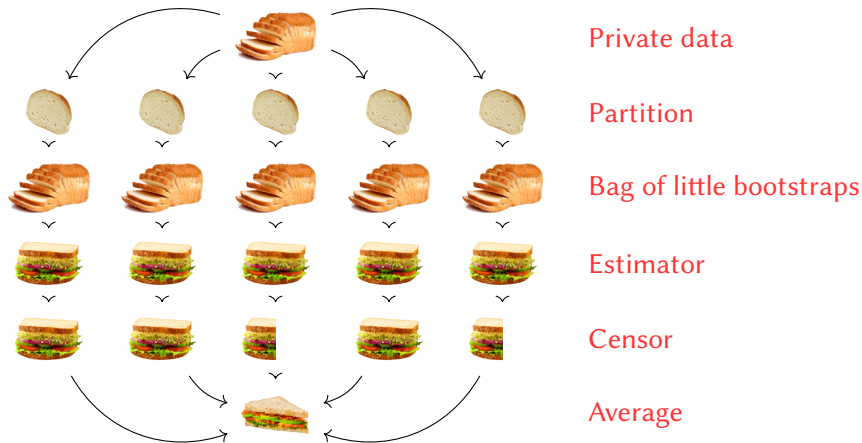
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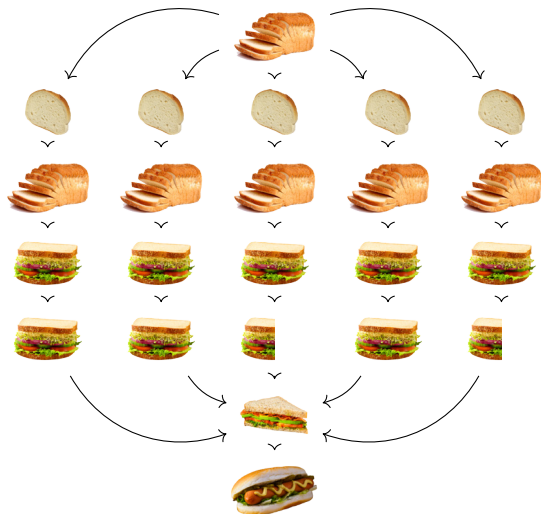
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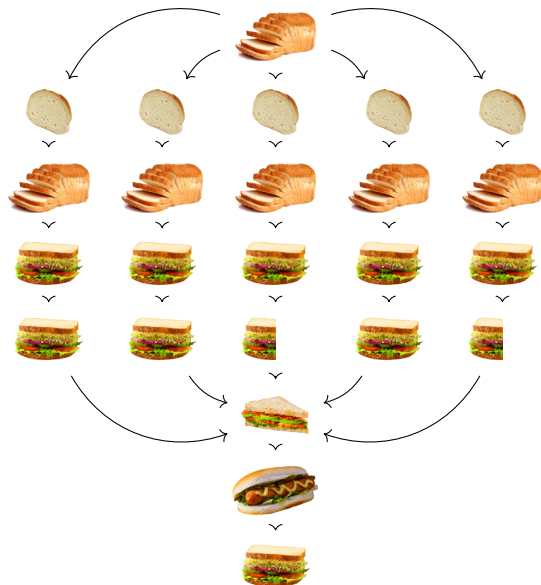
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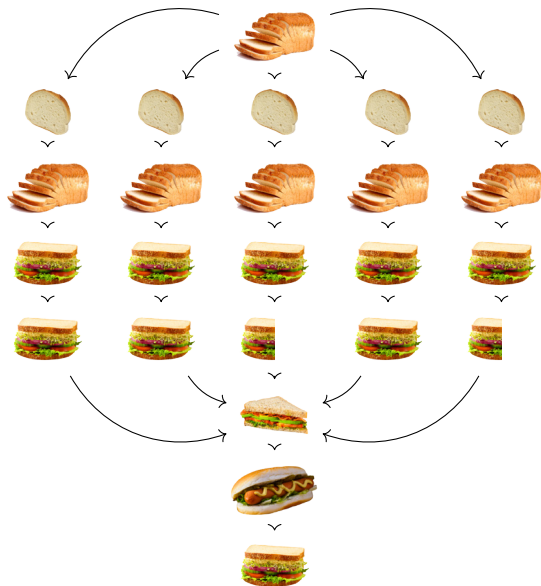
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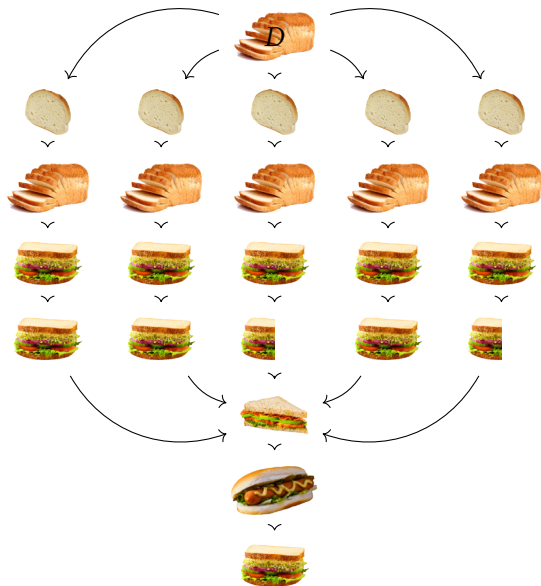
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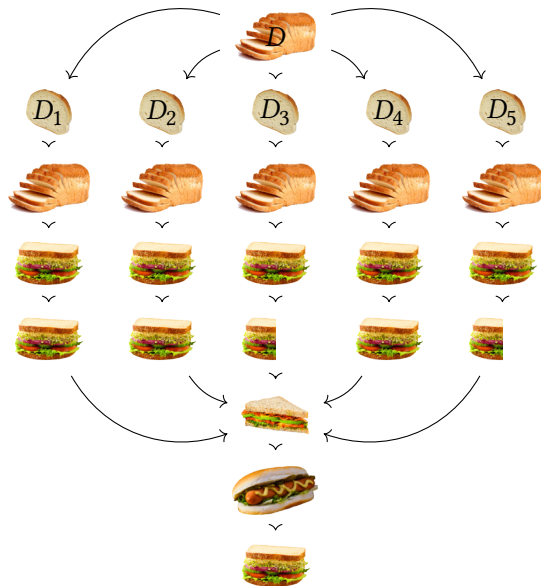
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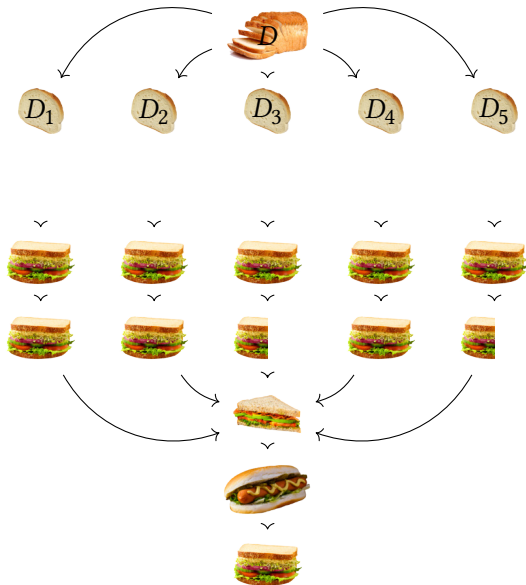
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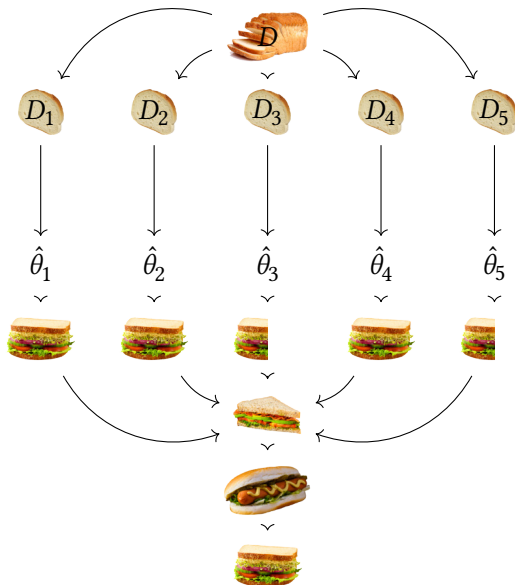
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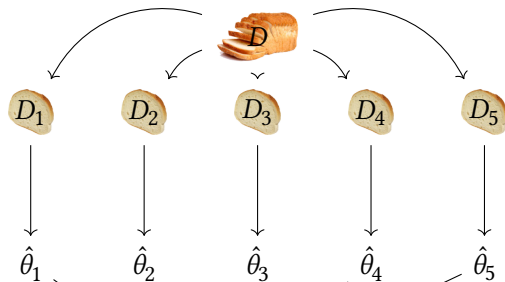
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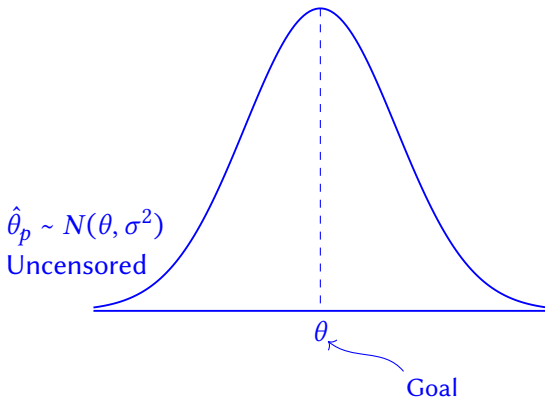
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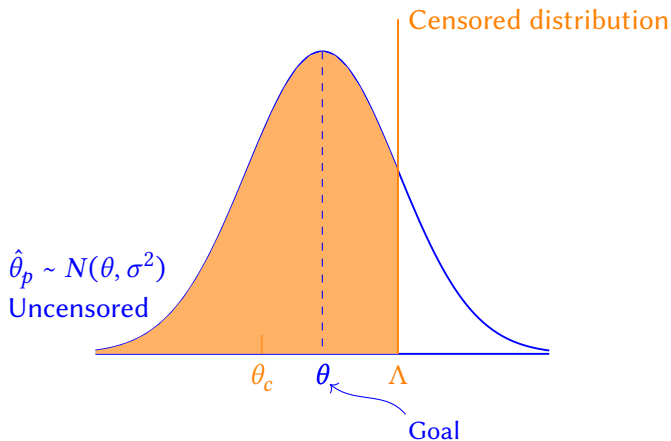
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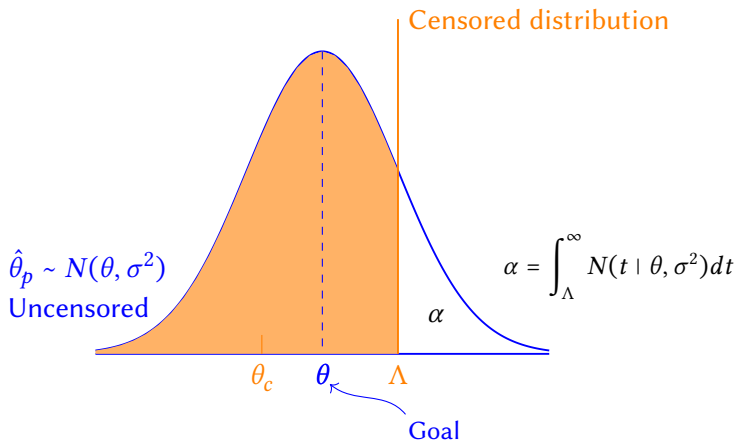
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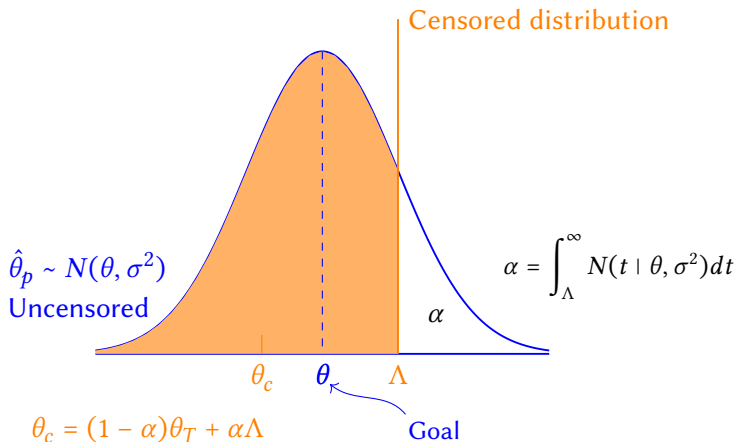
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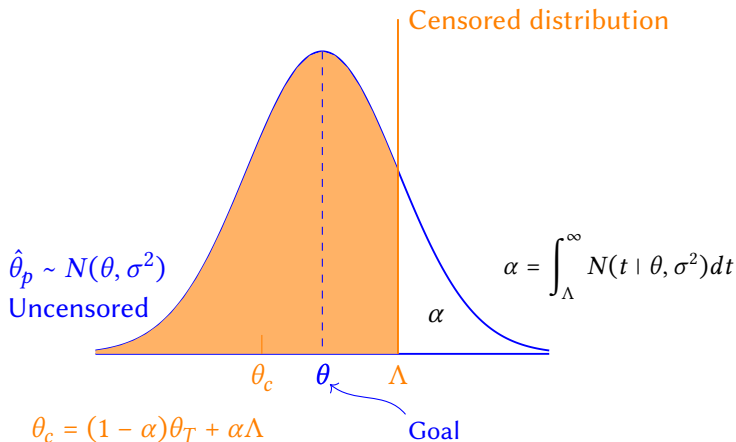
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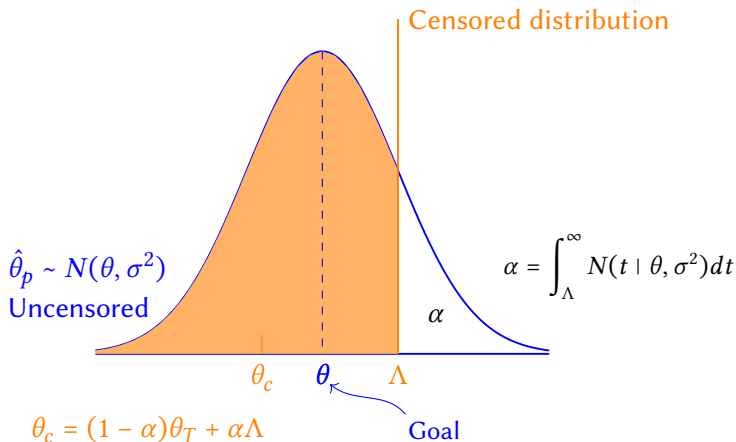


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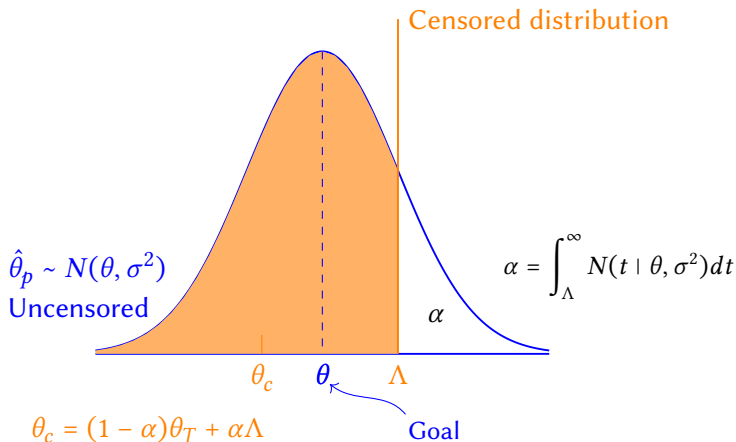
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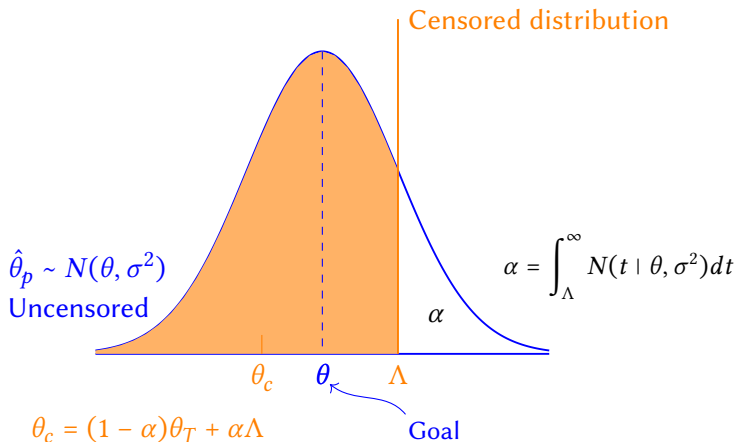
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Disclose: $\hat{\theta}^{\text{dp}}, \hat{\alpha}^{\text{dp}}$

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Functions of disclosed params

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Solving Political Problems Technologically

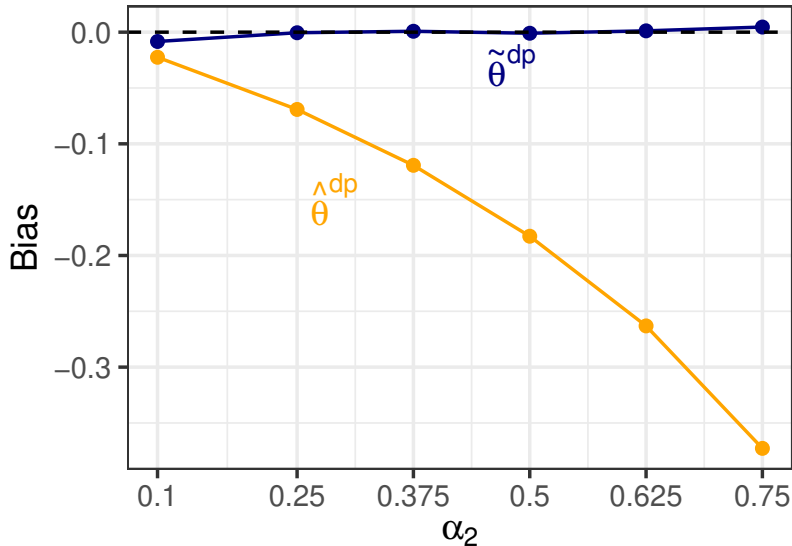
Differential Privacy & Inferential Validity

A General Purpose, Statistically Valid DP Algorithm

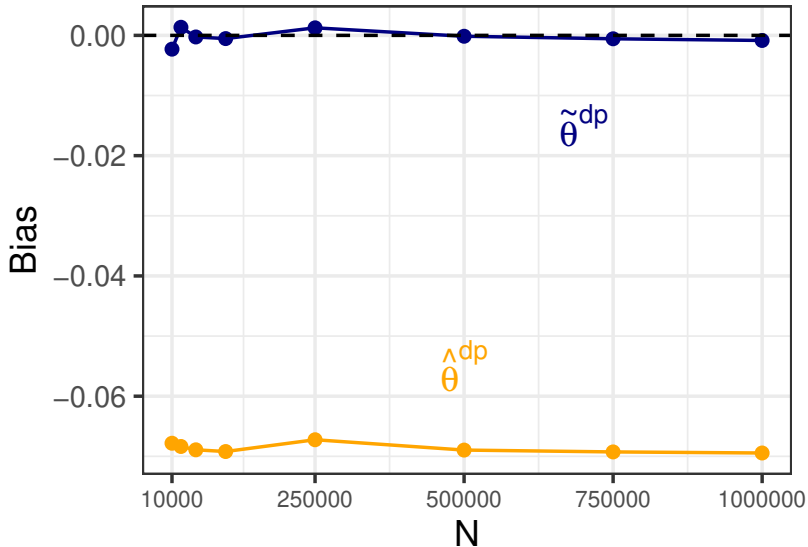
The Algorithm in Practice

Simulations: Finite Sample Evaluation

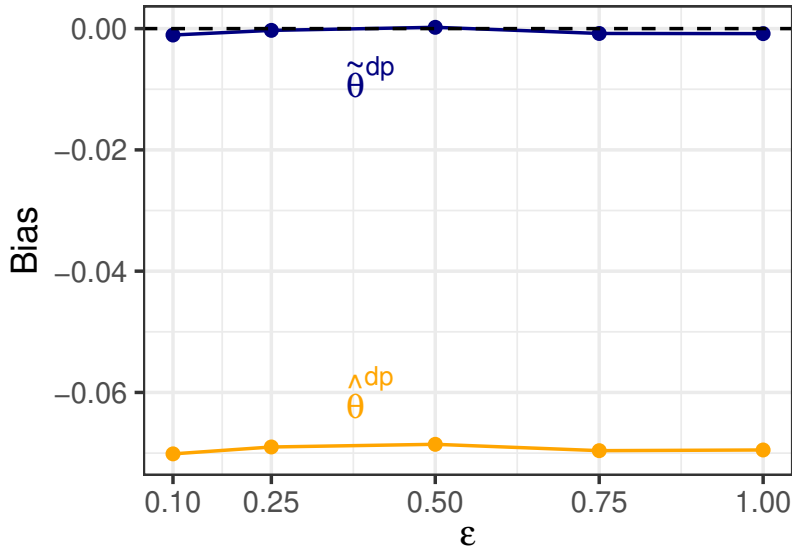
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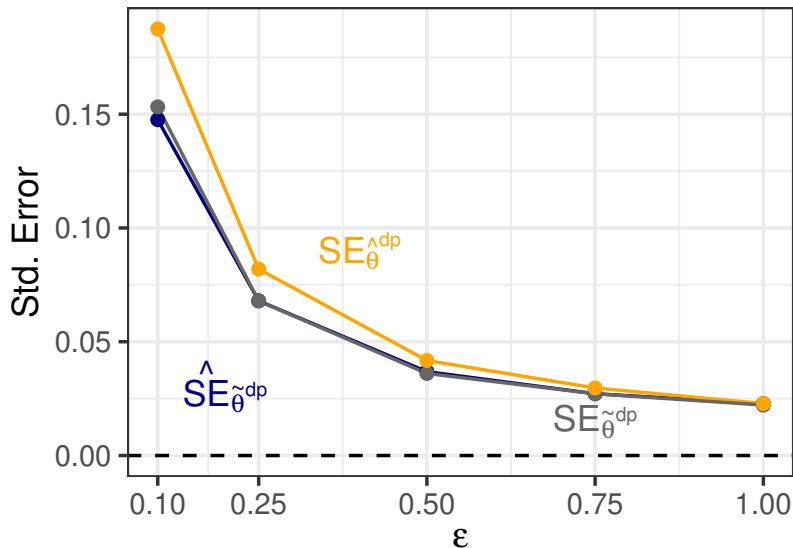
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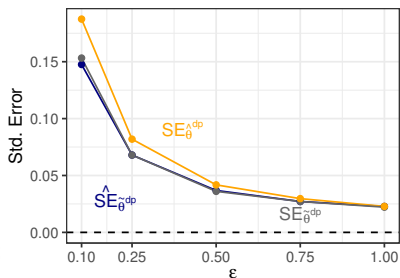
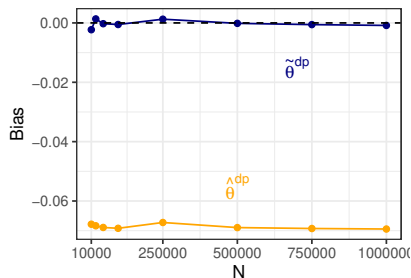
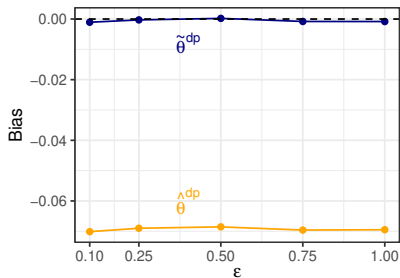
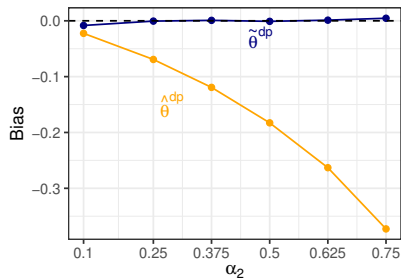
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For more information



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